



**Verified Carbon
Standard**

GROUPED PROJECTS FOR LAOS IMPROVED COOKSTOVE

Document Prepared by:

Sustainability Investment Promotion and Development

Joint Stock Company

(SIPCO)

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The intention of the grouped project activity involves the installation of energy efficient Improved Cooking Stove (ICS) in the households (hereafter “users”) of Lao People's Democratic Republic (here after “Laos”) for cooking households. For this reason, this proposed project activity aims at reducing the woodfuel consumption of traditional stove users by distributing ICS to households. By reducing the consumption of woodfuel, it can contribute to improving public health problems. Furthermore, this proposed project can contribute to Laos's climate change respond and sustainable development.

The project activity targets households with no previous ICS access. The geographical boundary is in Laos. ICSs are distributed free of cost to users. Sustainability Investment Promotion and Development Joint Stock Company (SIPCO) financed the implementation of the project in full.

The Project activity will promote the dissemination of energy efficient ICS. This will in turn reduce deforestation and degradation of forests in Laos through participation of the people in adopting energy efficient stoves. This will also contribute to improvement in quality of lives of the people from Laos through reduction of drudgery, time and money spent on fuel wood collection and through improvement of indoor air pollution. Globally, the project will contribute by reducing emission of GHG into the atmosphere.

The scenario existing prior to the implementation of the project is the actual baseline scenario which is equal to the existing common practice by the households i.e. in absence of the project activity the equivalent amount of thermal energy generated using firewood in three stone open fires to meet the cooking energy demand. The grouped project will distribute free of 600,000 ICSs to households who using traditional cookstove for cooking meal. All the project instances to be included in this grouped project will be from within Laos only.

PP signed distribution contract with Lao Red Cross (LRC) on 27/10/2022. Lao Red Cross is a social organization of Lao PDR that conducts humanitarian work independently under the Constitution of Lao PDR. Its role is to implement humanitarian work that engages participation of society, mainly in the areas of basic relief assistance, mobilization of blood donation and safe blood supply, primary health care promotion, first aid, emergency rescue, communication and fundraising and volunteer development, restoring family links, and international relations and cooperation, in collaboration with relevant organizations in order to develop and promote humanitarian values with no discrimination.

The distribution of ICSs in this grouped project is divided into 02 stages:

- Stage 1: From distribution signed date to December 2023, with the distribution of 300,000 ICSs.
- Stage 2: After the completion of Stage 1, with the distribution of 300,000 ICSs.

The estimates of annual average GHG emission reductions of the grouped project activity is 3,588,237 tCO₂e/annum. The project proponent aims to disseminate 8,000 ICSs in the first project activity instance of this grouped project. The estimated annual emission reductions from project activity instance 1 is 50,532tCO₂e/year

1.2 Sectoral Scope and Project Type

The project activity under consideration is a grouped project activity. The project activity instances as part of the grouped project will have following sectoral scope, project type and methodologies:

Sectoral Scope applicable to project: 03 – Energy Demand

Project type: Type II – Energy efficiency improvement project

Methodology: VMR0006: Methodology for installation of high efficiency firewood cookstoves.

1.3 Project Eligibility

Project activity involves in energy efficient cook stove distribution which falls under the category of efficiency improvements in thermal applications and reduces the use and demand of fossil fuel and non-renewable biomass that would have been used to cook meals. The baseline & monitoring methodology applied for this project is “Methodology for installation of high efficiency firewood cookstoves”, version 1.1. The host country of the project is Laos. Therefore, project activity does not fall under the excluded list of projects as prescribed under the table 1 of section 2.1.1. of the VCS Standard version 04.4.

1.4 Project Design

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

For the inclusion of new project activity instance. The project proponent shall ensure that it meets the eligibility criteria below:

No.	Eligibility criterion - Category	How the new project activity instances to comply
1	Meet the applicability conditions set out in the methodology applied to the project.	The new project activity instance will meet all applicability conditions of VMR0006: "Methodology for installation of high efficiency firewood cookstoves", version 1.1
2	Use the technologies or measures specified in the project description	Each Instance under this Grouped project will consist of the distribution of Ecozen cookstove with a thermal efficiency of 38.44% to household users cooking with non-renewable biomass in the baseline scenario. The instance consists of replacement of conventional firewood cookstoves for biomass fired ICS as defined in the Grouped project.
3	Apply the technologies or measures in the same manner as specified in the project description	Each instance under this grouped project shall consist of the distribution of Ecozen cookstove with efficiency improvements in thermal applications of non-renewable biomass. The Ecozen cookstove have a thermal efficiency of 38.44%
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area..	The new project activity instances will be installed within Laos only and subject to the same baseline scenario determined in section 3.4

5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new project activity instances will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no mandated government programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology. Step 3: Project method This step is not applicable under project activity because the stoves are provided at zero cost to the households (end user).
6	Conditions that avoid double counting of emission reductions.	The new project activity instances will: 1. Apply a unique system (i.e. conformity letter) for ICS in each project instance, assigning a unique ID to each ICS and allowing to clearly identify each ICS and to which project instance it belongs. 2. Each ICS installed will include project instance assignment and will have corresponding end user details (i.e. name, address). This data will be recorded as part of the beneficiary agreement and Project database. The CME has an agreement in place with residents who use ICS in the project instance in which the owner transfers the rights to the emissions reductions exclusively to the CME as part of the Carbon Rights Waiver within the Beneficiary agreement

7	Capacity limit	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified. 2) Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit
8	Target group	Each project instance will target end users which includes households who use traditional cookstove to cook meal.

1.5 Project Proponent

Organization name	Sustainability Investment Promotion and Development Joint Stock Company
Contact person	Ms. Nguyen Thi Hong Hanh
Title	Director
Address	6th Floor, Hop Long Building, No.1, 59 Lang Ha Street, Ba Dinh District, Hanoi, Viet Nam
Telephone	+84 243 519 0955
Email	info@carbonvietnam.com

1.6 Other Entities Involved in the Project

Not Applicable.

1.7 Ownership

The Project Ownership is with “Sustainability Investment Promotion and Development Joint Stock Company.” who is also the Coordinating & Managing Entity of the grouped project, the ownership of carbon credits shall be demonstrated through following indicative information

- The Project ownership purchases equipment directly from the manufacturer Viet Charcoal Production Company Limited. The cookstove purchased contract shall be signed between the two parties for each project activity (project instance) in this grouped projects. Every contract will be attached with a statement or/and a beneficial agreement between the project owner (SIPCO) and the cookstove manufacturer, the statement or/and beneficial agreement is to confirm that all carbon credits right from the purchased cookstoves for the project activity shall be remained to the project owner without affected result with cookstove manufacturer.
- On the other hand, although ICS beneficiaries possess ownership of the equipment, before installation of each project instance, beneficiaries enter into a beneficiary agreement with SIPCO and prove that SIPCO’s proof of ownership of the credits, through the following declaration “Irrevocably transfer all rights to any Carbon Credits that may be generated by the operation of the improved cookstove to SIPCO, and to take all reasonable steps to affect such transfer; I irrevocably waive any rights to Carbon Credit in exchange for the free supply of the improved cookstove”.

1.8 Project Start Date

The starting date is on 10/03/2023.

As per section 3.7 of the VCS Standard (v4.4), The project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals.

The project activity is a non-AFOLU project and first cookstove under the project has been operating from 10/03/2023. Thus, the project start date is on 10/03/2023.

1.9 Project Crediting Period

Fixed crediting period has been chosen.

Start date of crediting period : 10/03/2023

End date of crediting period : 09/03/2033

Total number of years : 10 years, 00 months.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☒ >1,000,000 tCO₂e/year

The estimated volume of emission reduction from the grouped project is 3,588,237 tCO₂e/year which is larger than 300,000 tonnes of CO₂e per year. Therefore, in accordance with the para 3.9.1 of the VCS Standard, v4.4 the scale of the project activity is under “Larger Project” category.

Project Scale	
Project	
Large project	x

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	2,017,198
Year 2	3,978,615
Year 3	3,923,391
Year 4	3,868,719
Year 5	3,814,595
Year 6	3,761,011
Year 7	3,707,963
Year 8	3,655,446
Year 9	3,603,454
Year 10	3,551,982
Total estimated ERs	35,882,374

Total number of crediting years	10
Average annual ERs	3,588,237

1.11 Description of the Project Activity

The purpose of this project is the dissemination of improved cooking stoves (ICS) in Laos. The project activity help to reduce the woodfuel consumption of traditional stove users by distributing improved cook stoves to households. By reducing the consumption of woodfuel, it can contribute to improving public health problems. Furthermore, this project activity can contribute to Laos's climate change respond and sustainable development.

To reduce deforestation and the emission of greenhouse gas in Laos, ICSs that will lead to use less woodfuel, will be distributed to the households and SMEs which mostly use traditional and/or conventional cookstoves with the woodfuel. In relation to the type of ICSs, which has at least 25% efficiency, will be distributed to a target group (Household). Over the long-term, this project activity is anticipated to reduce deforestation and the emission of greenhouse gas caused by using the burning of woodfuel.

In line with the applied methodology VMR0006 “Methodology for installation of High efficiency firewood cookstoves”, version 1.1, it is assumed that in the absence of the project, the baseline scenario is the continued use of non – renewable wood fuel by target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. Therefore, by reducing the total amount of fuel required for cooking, the replacement of traditional stoves by ICS reduces the amount of GHG emitted into the atmosphere. Emission Reductions are calculated following the methodology on the basis of the mass of non-renewable woody biomass saved by the ICSs.

Technology

Emission reductions from the ICS units will be calculated using the latest approved methodology VMR0006. “Methodology for installation of High efficiency firewood cookstoves”, version 1.1; hence falling under the Type II project activity – Energy efficiency improvement projects.

A typical project activity instance will replace traditional firewood stoves with higher efficiency ICS that are more efficient than traditional fire wood stoves as they reduce heat loss and improve heat transfer and/or combustion efficiency. The ICSs are fuel efficient, resulting in a decrease in fuel use in comparison to conventional pre-project stoves while also reducing particulate matter and carbon emissions. Design considerations have also streamlined assembly and construction to reduce cost and production times.

The ICS models are fuel efficient, resulting in a decrease in fuel use in comparison to conventional pre-project stoves while also reducing particulate matter and carbon emissions.

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These stoves deliver the same energy output as thermal energy supplied in the baseline scenario with traditional stone fire.

The ICSs are manufactured in Viet Nam. The manufacturer has had previous experience with the manufacture of cookstove and has a trained workforce for the production of the cookstove. The cookstove named Ecozen with characteristics below:

As per water boiling test conducted by GERES Biomass and Energy Laboratory (G-BEL), the thermal efficiency of the Ecozen ICS is projected to be 38.44%. The ICS has an expected lifespan of at least 10 years.

The Ecozen stove allows for significant savings of firewood compared to traditional stoves and is especially suitable for the cooking habits of the people in the target group. Its design ensures efficient combustion of fuel and cleaner fire and therefore uses considerably less fuel as compare to traditional stoves. This means speedy cooking, time and fuel savings, cleaner pans, kitchen walls and indoor atmosphere.



The Ecozen cookstove is being produced and planned for distribution for project activities instance in the grouped project. Other cookstove models will be added later if there are plans to distribute them.

The project promotes energy efficient improved cooking stove (ICS) which directly reduces fuelwood consumption, which are non-renewable biomass having higher emission factor. As an associated benefit, the project directly helps reducing the rate of deforestation in the host country.

1.12 Project Location

The grouped project will be hosted by Laos. The grouped project covers the whole of Laos.

Latitudes: 19.8563° N

Longitudes: 102.4955 E

The detail of coordination of Laos shall be provided separately in KML file.



Figure 1: Map of Laos

1.13 Conditions Prior to Project Initiation

The project reduces the use and demand for fossil fuels and non-renewable biomass that would have been used in the traditional stove to achieve the same output (i.e. cooking daily meals) with the ICS. This directly leads to reduced GHG emissions. Therefore, the project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. Prior to project initiation households are using traditional cooking stoves or three stone fires.

Laos 2020 population is estimated at 7,275,560 people at mid-year according to UN data, in which the population in urban is 2,640,299 people and in rural is 4,635,257 people¹. The high proportion of non-renewable biomass demonstrates that typical household cooking meal contributes to deforestation and threatens biodiversity. In Laos alone, 77.4% of rural households and 98%² in the poorest quintile use firewood for cooking with significant portions of the population using unimproved cooking stoves – because the use of electric stove and LPG/cooking gas stoves is very limited that accounts for only 6.5%³ of households and among most common uses cookstove, none of them can meet requirements of an improved cookstoves. The use of wood for burning directly leads to reduced greenhouse gas emissions.

The wood collected or harvested to fire traditional stoves. According to the approved methodology, in the absence of the project activity, the baseline scenario would be the use of wood fuel for the community to meet its energy need. Therefore, by reducing non-renewable biomass consumption (i.e. fuel wood), the grouped project is reducing anthropogenic GHG emissions.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project activities under the grouped project are voluntarily coordinated action Sustainability Investment Promotion and Development Joint Stock Company. There is no mandatory law or requirement in Lao People's Democratic Republic to foster the dissemination of ICSs.

Further to mention that in terms of local laws, statutes and regulatory frameworks etc. specific to the project type, there is no specific laws/policies available for cookstove projects. The distribution and implementation of ICS technology is not subject to any environmental impact assessment in accordance with the EIA (Environmental Impact Assessment) Guidelines - MONRE of Laos⁴. Category 2 in Appendix 1 of this guideline lists activities/projects that must prepare the environmental report. Thus, the distribution of ICS in households is not included in the list.

¹ <https://www.worldometers.info/world-population/laos-population/>

² The Lao Social Indicator Survey II in 2017 by Lao Statistics Bureau, Table TC.4.2: Primary reliance on solid fuel for cooking

³ The Lao Social Indicator Survey II in 2017 by Lao Statistics Bureau, Table TC.4.1: Primary reliance on clean fuels and technologies for cooking, page 187.

⁴ Environmental Impact Assessment Guidelines, prepared by Environmental Management Support Programme (EMSP), Oct.2012, available from:
http://monre.myqnapcloud.com/2017/emsp/images/doc/eia_guidelines20121001.pdf

Thus, it can be concluded that there is no applicable local or regional law and regulatory framework in the host country with relation to cookstove projects

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

Not Applicable

1.15.2 Projects Rejected by Other GHG Programs

Not Applicable

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

The project will not generate any other form of GHG-related environmental credit for GHG emission reductions claimed under the VCS program. The serial ID number of each ICS distributed under the project is recorded, and each unique serial ID is assigned to a single user whose details are recorded at the time of stove purchase. A searchable database containing these details enables cross checking.

The project activity under VCS shall not claim emission reductions for the same period that has been/will be covered under other GHG Programs. Thus, there will not be any possibility of double accounting of emission reductions

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

If yes, provide the name of the other program(s) under which the project has sought or received another form of GHG-related credit.

Supply Chain (Scope 3) Emissions

Have the owner(s) or retailer(s) of the impacted goods and services⁵ posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with Ecozen Cookstove, manufactured by Viet Charcoal Production Company Limited in Di Su Commune, My Hao District, Hung Yen Province, Vietnam, distributed by Sustainability Investment Promotion and Development Joint Stock Company (SIPCO) under their project activity titled “Grouped projects for Laos Improved cookstove” (VCS ID: 2924) since the project’s start date?

☐ Yes ☒ No

Viet Charcoal Production Company Limited is in charge of manufacturing improved cookstoves for SIPCO's greenhouse gas emission reduction projects. The company received an email from SIPCO requesting a public statement for scope 3 emission of the project. However, Viet Charcoal's website is only used to introduce their products to partners, and does not publish information on their partners' projects to avoid misunderstanding. Therefore, all public statements for supply chain of scope 3 emission will be announced directly by the project proponents on their websites instead.

Has the project proponent posted a public statement saying, “VCUs may be issued for the greenhouse gas emission reductions and removals associated with Ecozen Cookstove, manufactured by Viet Charcoal Production Company Limited in Di Su Commune, My Hao District, Hung Yen Province, Vietnam, distributed by Sustainability Investment Promotion and Development Joint Stock Company (SIPCO) under their project activity titled “Grouped projects for Laos Improved Cookstove” (VCS ID: 2924) (Detail in Appendix 1 of this report)

☒ Yes ☐ No

Sustainability Investment Promotion and Development Joint Stock Company (SIPCO) have posted the public statement on their website.

Have the producer(s) or retailer(s) of the impacted good or service been notified of the project and the potential risk of Scope 3 emissions double claiming via email?

☒ Yes ☐ No

SIPCO have sent the notification of the project and the potential risk of Scope 3 emission double claiming to Viet Charcoal Production Company Limited via email (Detail in Appendix 1 of this report).

⁵ Impacted goods and services are all goods and services directly impacted by the technologies and measures specified as project activities in the project description. Please see the VCS Program document *VCS Program Definitions* for additional information.

1.17 Sustainable Development Contributions

The project activity helps in achieving following co-benefits, which will contribute in sustainable development in host country.

A) Environment well-being:

- The project activity will contribute to the reduction in GHG emission due to reduction in use of non-renewable biomass and it can ensure natural recovery of forests and/or reforestation.
- The project activity will lead to reduction in indoor air pollutants from wood smoke and avoid smoke related health disorders.
- The reduction in non-renewable biomass consumption for cooking and boiling water through efficient use leads to improved ecological balance.
- The project activity will support fire prevention in the household kitchen.

B) Socio and Economic Benefits:

- Implementation of the project activity results in employment opportunities for people involved with installation, maintenance and distribution of these household biomass based cookers.
- The project activity will improve the status of living of people in households cooking since the community level implementation of units of these kinds can be carried out at considerably lower costs. There will be less dependence on firewood and its expenses and time associated with firewood buying and collection.
- Business development component of the project activity has resulted in the enhancement of capacity development and technical knowhow for moderately educated people in the community through workshops, seminars and training programs.
- The project will alleviate the work burden of women and children as they have to spend less time to collect firewood for cooking.
- Community health issues such as chronic lung diseases, acute respiratory infections, cataracts, blindness, and adverse effects on pregnancy will be decreased due to improvement of indoor air quality.

C) Technological benefit

The introduction of locally manufactured technology with improved energy efficiency helps in technological self-reliance in the area.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable, as project activity has been design to include only new improve cookstoves. No used improved cookstoves will be transferred or deployed from outside the geographical project boundary to the project activity.

Additionally, the project activity applies Methodology VMR0006. "Methodology for installation of High efficiency firewood cookstoves", version 1.1. According to the section 8.3 of the methodology, the default net to gross adjustment factor of 0.95 has been applied to account.

Commercially Sensitive Information

Not Applicable.

Further Information

Not Applicable.

2 SAFEGUARDS

2.1 No Net Harm

Greenhouse gas reductions: Project implementation will result in GHG reductions because it will reduce the consumption of non-renewable biomass in countries where the biomass harvested for fuel use is typically non-renewable.

Air quality: Users (especially women and children) will be exposed to fewer air pollutants through reduced emission of not only CO₂, but also carbon monoxide and particulate matter. Air pollution from cooking with solid fuel is a key risk factor for childhood pneumonia as well as many other respiratory, cardiovascular and ocular diseases.

Biodiversity: will be improved as the programme reduces pressure on remaining forest reserves in Laos. No negative impacts can be identified.

2.2 Local Stakeholder Consultation

An invitation letter was sent to related local stakeholders including agencies, Lao Red Cross, Ministries and NGOs, project developers, manufacturers, local vendors and households in 8th August 2022 in order to inform about the local stakeholder consultation meeting conducting in 18th August 2022 at the meeting room in Vientiane. We also send email and made the phone call to all invited local stakeholders for clearer clarification on the project and getting the confirmation of the meeting attendance. Stakeholders were identified as those whose activities directly or indirectly impact the project, and those who were to be impacted by the project activities. The invitations were welcoming written comments as well as physical attendance.

In the meeting, Ms. Phonedavanh Sanbounleuxay, Deputy Head of International Relations Division, introduced the participants of the stakeholders meeting “Grouped Projects for Laos Improved Cookstove”.

Ms HO THI QUY, the representative of the sponsor entity, made a presentation on SIPCO, the strong relationship between Vietnam and Laos. Ms. Ho Thi Quy appreciated the presence and support of Lao Red Cross as well as Ministries of Laos and other associations, and looked forward to further cooperation between Parties. Ms. Ho Thi Quy also mentioned

the economic, social and environmental impacts of the Project to the local community. The project goal is free distribution of improved cookstoves (ICSs) to households using traditional cookstoves in Laos.

Each attendee was handed out feedback form, project description presentation therefore, their comment could be collected directly in the meeting or in other hand sent to CME via email within 15 days after the date of the local stakeholder meeting.

Currently, the project received all supportive comments from local stakeholders without any negative one.

Consultation will continue throughout the project lifetime, and will continue to influence project design. Adaptations will be made if deemed necessary

Mechanism for ongoing communication

SIPCO engages with Lao Red Cross through annual reports, records and SIPCO's social media communication. Lao Red Cross are large organizations at all administrative levels (Commune, District, Province, State).

- Annual Reports includes information about the operation throughout the year. Lao Red Cross member at commune and ward levels regularly come to exchange information with Beneficiaries and report to the District's level and the District's level will report it province's level. The province's Red Cross has responsibly for summarizing and reporting to SIPCO. If these Beneficiaries have any comment in using cookstove, the Red Cross will report immediately to SIPCO.
- Records will send to project donors about 2 times a year. The records include updates on current projects, community stories to share, fund raising appeals and much more.
- Social media communications are crucial to stay engaged with a variety of stakeholders. Below mentioned mechanism will be used by stakeholders to communicate their grievances with the Project owner:
 - Communication via email info@carbonvietnam.com
 - By informing Red Cross at levels.

SIPCO will continue to implement these informal ongoing consultation procedures, alongside formalized consultation via annual our survey.

2.3 Environmental Impact

The implementation of the grouped project does not impose any severe impacts on the ecological system in the surrounding areas. This project activity will help in reducing the consumption of fuel wood by efficient fuel combustion thereby reducing deforestation and reducing indoor air pollution. The use of efficient Cooking Stove technology will reduce adverse environmental and social impacts associated with the use of non-renewable

biomass. The reduced consumption of the stated baseline fuel will have sustainable benefit for all household's traditional cook stoves users included in the project activity. The implementation of the project will reduce workloads involved in fuel collection, and by reducing indoor air pollution, will reduce the risk of respiratory diseases, especially for women and children. Efficient or improved Cooking stoves will also contribute to environment protection by reducing biomass consumption and hence greenhouse gas emissions, mainly CO₂, when the combusted biomass originates from non-renewable stocks

Further to mention that in terms of local laws, statutes and regulatory frameworks etc. specific to the project type, there is no specific laws/policies available for cookstove projects. The distribution and implementation of ICS technology is not subject to any environmental impact assessment in accordance with the EIA (Environmental Impact Assessment) Guidelines - MONRE of Laos⁶. Category 2 in Appendix 1 of this guideline lists activities/projects that must prepare the environmental report. Thus, the distribution of ICS in households is not included in the list. Therefore, the Environmental Impact Assessment (EIA) is not required for the project.

2.4 Public Comments

The project activity received 03 comments during the public comment period from 22 April 2022 to 21 May 2022. Detail below:

Comment 1

Date: 01/05/2022

Comment:

Please provide a Certificate for the thermal efficiency of the Ecozen ICS is 38.44%. The Certificate for that ICS has an expected lifespan of at least 10 years. The product cannot have at least 10 years lifespan and thermal efficiency is 38,44%. This is a scam. Request Laos Government Agencies/MONRE of Laos to check the quality carefully.

Response by Project Participant

The efficiency test for the Ecozen model introduced in the grouped projects are 38.44%, the efficiency were conducted by a thermal efficiency specialized Laboratory, GERES Biomass and Energy Laboratory (G-BEL), recommended by the Clean Cooking Alliance⁷ (United Nations run organization), the full test report and certification have been provided to the VVB for verifying.

⁶ Environmental Impact Assessment Guidelines, prepared by Environmental Management Support Programme (EMSP), Oct.2012, available from:
http://monre.myqnapcloud.com/2017/emsp/images/doc/eia_guidelines20121001.pdf

⁷ <https://cleancooking.org/sector-directory/geres-biomass-energy-lab-g-bel>

The lifespan of the Ecozen cookstove model have been confirmed by the manufacturer as the lifespan test report provided to VVB which is compliance with the applied methodology.

For the registered number of WIPO for the products, the ECOZEN cookstove has been certified for its exclusive industrial design by the Intellectual Property Office of Vietnam No. 33069 dated 01/07/2021, the certificate has been provided to the VVB (The Intellectual Property Office of Vietnam is a member of WIPO⁸).

In other hand, for the cross check with recently listing cookstove projects for comparison of efficiency and lifespan, the following results have been found:

No.	Registry ID	Host country	Cookstove Model	Efficiency	Life time	Type of cookstove
1	2533	India	Green Jumbo	38%	7 years	Steel stove
2	2505	South Africa	Zama Zama Rocket stove	38.3%	5 years	Steel stove
3	2473	India	Agneekaa Eco Mini Stove	32.19%	7 years	Steel stove
4	2523	Viet Nam	Kuniokoa	48%	10 years	Steel stove
5	2409	Cambodia	SSM-S32-13	38.5%	10 years	Steel stove

Conclusion: The lifetime and thermal efficiency Ecozen cookstoves which are used in the grouped project are reasonable.

Comment 2

Date: 03/05/2022

Comment:

Dear Verra Registry,

The grouped project is related to the distribution of cookstoves, however, section 1.14 refers to water project.

Response by Project Participant

The grouped project is related to the distribution of cookstoves so section 1.14 has been revised accordingly.

Comment 3

Date: 03/05/2022

⁸ <https://ipvietnam.gov.vn/wipo-ip-hub>

Comment:

Provide LSC report which is organized with the participation of local authorities, households, NGOs, related parties, and representatives of Laos government agencies to ensure the project is allowed to operate in accordance with the laws of Laos Government. How comments are considered and evaluated?

Response by project participant

In 18th April 2022 at meeting room in Vientiane city, the representatives of the sponsor entity of Sustainability Investment Promotion and Development Joint Stock Company (SIPCO) together with the project implementation support partner – Lao Red Cross (LSC) holds a meeting with the stakeholders to collect their comments on the economic, social and environmental impacts of projects “Grouped projects for Laos Improved Cookstove”. The meeting with participation of the representatives of agencies, Ministries such as Lao Red Cross, Ministry of Natural Resource and Environment, Ministry of Foreign Affairs, Ministry of Labor, Invalids and Social Affairs... , NGOs including Swiss Red Cross, Gender Development Association (GDA)...and households. All of them strongly advocated for the construction of the project with the contents as presented by the Sponsor. All comments received are supportive of the project and hope the project will be implemented soon. The detail about LSC report and list of participants has been provided to VVB for verifying.

2.5 AFOLU-Specific Safeguards

The grouped project is a non-AFOLU project activity, hence not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VMR0006. “Methodology for installation of High efficiency firewood cookstoves”, version 1.1

This grouped project also refers to the approved versions of the following tools, methodology and methodologies:

- Tool 30 – “Calculation of the fraction of non-renewable biomass”; version 04.0
- AMS-I.E: “Switch from non-renewable biomass for thermal applications by the user”, version 12.0
- Guidelines for sampling and surveys for CDM project activities and programs of activities, version 04.0

- Standard Sampling and surveys for CDM project activities and programs of activities, version 09.0

3.2 Applicability of Methodology

Table 3.1. Applicability of applied Methodology

	VCS Methodology Requirement	Justification										
1	Project activities shall be implemented in domestic premises or in community-based kitchen.	The proposed project involves deployment of ICS only in households.										
2	The project stove shall have specified high power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	These Ecozen stoves planned to be installed under this project are single pot that have an efficiency of 38.44% as per the manufacturer’s specification.										
3	Both ‘Projects’ and ‘Large Projects’ can use the methodology	Estimated average annual emission reductions of the grouped project is 3,588,237tonne CO2 per year which larger than 300,000 tonnes CO2e per year. Therefore, proposed project qualifies the “Larger Projects” criteria.										
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	According to FAO data, the total forest cover in Laos has decreased since 1990. Global Forest Resources Assessment Country Report (2015) revealed that the total forest area has been declining, as indicated in the table below: Table: Trends in carbon stock in living forest biomass 1990 - 2015⁹ <table><tr><td>1990</td><td>2000</td><td>2005</td><td>2010</td><td>2015</td></tr><tr><td>1153</td><td>1130</td><td>1108</td><td>1086</td><td>1064</td></tr></table>	1990	2000	2005	2010	2015	1153	1130	1108	1086	1064
1990	2000	2005	2010	2015								
1153	1130	1108	1086	1064								

⁹ FAO forest Resource Assessment 2015 Global Tables.

		Based on the findings above, one can reasonably assume that Non-Renewable Biomass has been used before 1989 thereby hinting that fuelwood used across Laos comes from non-renewable sources.
5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Not applicable. The project use of biomass is sourced from non-renewable sources.
6	The PD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	Each ICS in this project will be identified by a unique combination of customer name and geographical location, as well as a serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project. In addition, the project has been cross-checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any

		other CDM project activity or voluntary project activity
7	The PD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	All users of ICSs sign a Beneficiary Agreement that cedes all rights to VRs to the project proponent. Therefore, only the project proponent is able to claim credit for emissions reductions from the project devices.

Table 3.2. Applicability of applied Tools

Tool	Applicability	The project
TOOL30: Calculation of the fraction of non-renewable biomass	This tool may be used by: (a) DNAs to submit region/country-specific default fNRB values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or (b) Project proponents to calculate project or PoA-specific fNRB values.	Complies. This tool was used by the project participants to calculate a project specific value for fNRB under option (b) of applicability condition 1
	For project or PoA specific fNRB values, project proponents shall assess the area where biomass is sourced and justify the selection of the area in CDM project design documents.	Complies. The area for the calculated value was selected to coincide with the Grouped Project's project boundary
AMS-I.E.: "Switch from non-renewable biomass for thermal applications by the user".	The use of this methodology in a project of activity under a grouped project is legitimate if the leakage is estimated and accounted for as per the relevant provisions of AMS-I.E under the section for grouped project.	Complies. Leakage emissions related to the non-renewable woody biomass saved by the project activity shall be calculated by applying B_y is multiplied by a net to gross adjustment factor of 0.95 to account for leakages. Therefore, AMS I.E is applied.

Guidelines for sampling and surveys for CDM project activities and programs of activities	This document describes common types of sampling approaches and includes a recommended outline for a sampling plan, recommended practices for unbiased estimates of sampled parameters and recommended evaluation criteria for designated operational entity (DOE) validation besides several best-practice examples covering large and small-scale project activities and PoAs. It also provides examples for checking the reliability of data collected through sample surveys	Complies. Representative sampling will be undertaken that is designed in line with the requirements of the applied methodology to monitor parameter $N_{y,i,j}$ and $B_{y=1, \text{ new, I, survey}}$
	Furthermore, it covers the following items: (a) Methods, if any, to deal with missed reliability targets without compromising conservative estimates for emission reduction (b) Best-practice examples for DOE validation/verification for sampling and surveys.	Complies. The single sampling plan covering a grouped project is undertaken applying 95/10 confidence/precision for the sample size calculation
Standard Sampling and surveys for CDM project activities and programs of activities	This document specifies the reliability requirements and describes appropriate sampling methods and what is expected to be provided in a sampling plan. The general requirements shall be applicable to both small-scale and large-scale CDM project activities and PoAs, with any requirements specified in the applied methodologies taking precedence. This document also provides sampling-related requirements pertaining to validation and verification,	Complies. Representative sampling will be undertaken that is designed in line with the requirements of the applied methodology to monitor parameter $N_{y,i,j}$ and $B_{y=1, \text{ new, I, survey}}$

	including the sampling approach to be undertaken by the validating/verifying DOEs.	
--	--	--

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/fossil fuel	CO ₂	Yes	Major emission source
		CH ₄	Yes	Minor emission source
		N ₂ O	Yes	Minor emission source
		Other	No	No other source identified
	Production & Transport of fuel	CO ₂	No	Minor emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass.	CO ₂	Yes	Major emission source
		CH ₄	Yes	Minor emission source
		N ₂ O	Yes	Minor emission source
		Other	No	No other source identified
	Production & Transport of fuel	CO ₂	No	Minor emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	No other source identified

A representation of baseline boundary and project boundary are given as below:

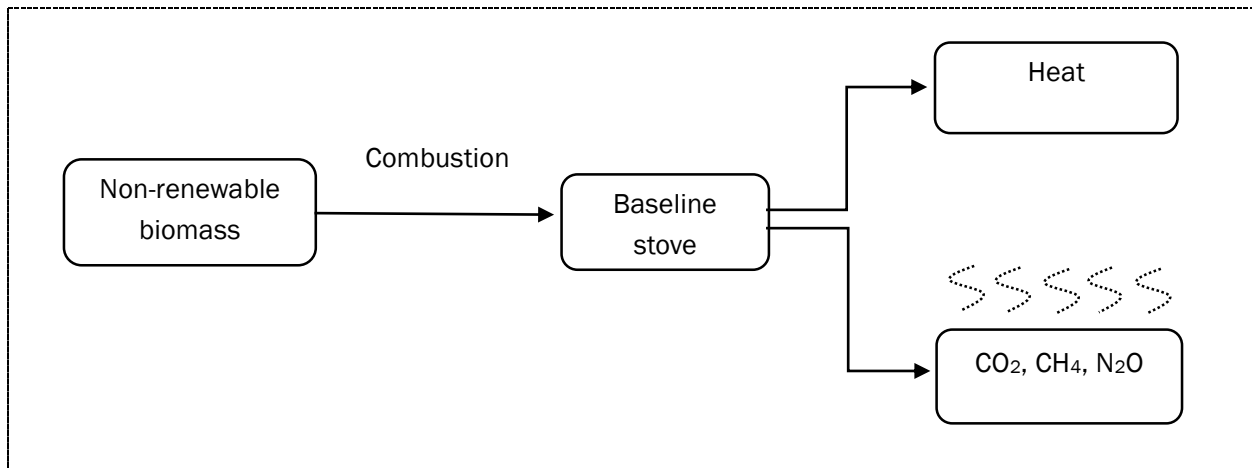


Figure 1: Baseline Boundary

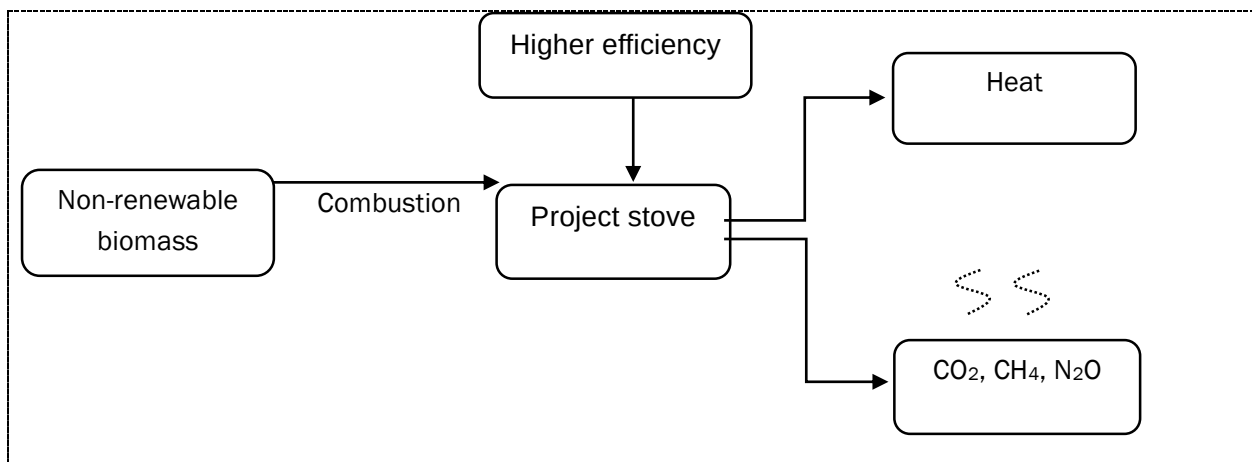


Figure 2: Project boundary

3.4 Baseline Scenario

In accordance with the applied methodology, the baseline scenario is continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. The baseline cookstoves are traditional cookstoves or three stone fires using firewood or conventional device with no improved combustion air supply or gas ventilation.

A baseline survey was conducted before distributing ICSs by SIPCO and Women Union at all level to confirm the type of baseline cookstove. The number of samples is calculated based on Guidelines for sampling and surveys for CDM project activities and programs of activities, v.04.0 and Standard Sampling and surveys for CDM project activities and

programs of activities, v.09.0, detail description in section 5.3. Survey results show that all selected households are using traditional stoves with 10% efficiency for cooking meal.

The project activities instance to be included in this grouped project activity should meet the baseline scenario as mention above.

3.5 Additionality

Methodology VMR0006 uses activity method for the demonstration of additionality.

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework. Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by SIPCO.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

1. The project distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits. The distribution of ICS require capital. The proposed project will distribute ICSs free of cost to users which means no money shall be charged to users for the distribution of ICSs; hence there is no financial return from the project other than revenue from the sale of GHG credits. As each ICS has costs associated with the purchase and operation of the equipment, this project will not occur in the absence of GHG credits revenue. The action is not financially viable without the support of revenues from the sale of GHG credits.

The grouped project will distribute 600,000 ICS free of cost to users i.e. households, hence there is no financial return from the programme other than revenue from sale of GHG credits. As each ICS has costs associated with the purchase and distribution of the ICSs, this grouped will not occur in absence of the GHG credits revenue. The action is not financially viable without the support of revenues from the sale of GHG credit. As the end-user doesn't benefit from a direct financial return and procuring ICSs requires capital, which is a barrier to rural consumers due to difficulties in accessing capital, a wide dissemination of ICSs in the Host Country is unlikely. The significant costs for this project activity, including ICS purchase cost, have been demonstrated to DOE.

2. The project is not implemented as part of government schemes or supported by multilateral funds.

Step 3: Project method

This step is not applicable under project activity because the stoves are provided at zero cost to the households (end user).

Conclusion: The grouped project is deemed as meeting steps above of the activity method and therefore, it is deemed additional. All new project activity instances will use the activity method for demonstration of additionally in monitoring period.

3.6 Methodology Deviations

Not applicable

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

In accordance with the applied methodology, the baseline scenario is continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity

The actual baseline scenario is the use of non-renewable biomass. The quantity of biomass used in absence of the project activity for target consumers will be determined at the Instance level. Assessments, information used in initial instances may be used in subsequent instances in lieu of conducting fresh assessments at each instance level.

The methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4

SIPCO coordinated with the Lao Red Cross to conduct baseline surveys for grouped projects in Laos. From the list of households selected by the Lao Red Cross, the PP calculates the sample size by establishing a 10% of error and confidence level up to 95% to determine the number of people used traditional cookstove, number of cookstove in each household, fuel in baseline.... The result of sample size is 384, we expected the response rate from the sampled households to be 90% then we would need to scale up this number accordingly. Thus we would decide to sample $384/0.9 = 430$ households. Applying the simple random sampling technical, the household numbers are randomly chosen from the required sample size as calculated by the CME. To ensure the survey result is representative of the entire population, the dissimilarity (such as village, individuals...) chosen randomly such that each

element (or unit) of the population has the same probability of being selected. The survey method has been use is face to face interview by questionnaires, result below:

Population:

Among 430 households being surveyed, the result is as below:

Table 01: Number of people in household

Number of people in household (person)	Number of household	Population in total households (person)
7	16	112
6	72	432
5	168	840
4	135	540
3	22	66
2	15	30
1	2	2
Total	430	2022

Therefore, the size of a household is estimated at $2022/430 = 4.7$ members.

Sex

Women take the important role and spend much time in collecting woods and cooking.

There are 54.65% (235 households) of the households being interviewed showing that the people using woods are mainly women, 18.84% (81 households) of the households being interviewed showing that the people using woods are men. The rest of 26.51% (114 households) assume that both women and men use woods for cooking. Assumed that the rate of women and men is approximate then the being shown above could be inverted as: 67.91% women, 32.09% men use woods for daily cooking.

Number of cookstove

As the result of the survey, 95.58% of the households (411 households) being interviewed are using traditional 3 stones cookstoves only for cooking, 4.42% (19 households) are using two types of cookstoves (traditional cookstove and other cookstove such as improved cookstove), no household answer that they are using other stoves only.

Woody fuel consuming

Among 430 households are randomly surveyed, all of households use woody fuel to cook, total woody fuel consumed by households is 1,521,720 kg/year. So, the average woody fuel consumed by a household at $1,521,720/430=3,538$ kg/household/year=3.538 tons/household/year.

Type of fuel

The survey team use the questionnaire of using fuel in the households. All of households use woods as the main fuel for cooking. Among 430 households being surveyed, none of them use gas, coal, oil and electric for cooking. 100% of households use wood fuel for cooking.

Moreover, before starting the distribution of improved cookstove in a new area, PP conducts a feasibility check (through baseline survey) through which the proportion of using traditional cookstove is being confirmed. In case that the proportion of other cookstove is significant, the area will be discarded. At the time of distribution, households are checked for eligibility. Households indicating as their baseline stove anything else than traditional stove are ineligible for the carbon programme. This shall be ensured before household receiving the cookstove as the households have to commit and sign the beneficiary agreement with SIPCO

4.2 Project Emissions

Refer to section 4.4 for the calculation of emission reductions resulting from the proposed project. There is no project emissions considered.

4.3 Leakage

As detailed in section 8.3 of the methodology VMR0006, Version 1.1, leakage shall be considered as default 0.95.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

Emission reductions are calculated as:

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Where:

- | | |
|---|---|
| i | Indices for the situation where more than one type/model of improved cookstove is introduced to replace three stone fire. |
| j | Indices for the situation where there is more than one batch of improved cookstove of type i |

ER_y	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	Emission reductions by improved cookstove of type i and batch j during year y in tCO ₂ e

Since the project activity instance involves the deployment of improved cookstove, the following equation is applicable as per VMR0006, Version 1.1:

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where:

$B_{y,savings,i,j}$	: Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
$f_{NRB,y}$	: Fraction of woody biomass that can be established as non-renewable biomass (f_{NRB})
$NCV_{woodfuel}$	: Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)
EF_{wf,CO_2}	: CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ)
$EF_{wf,non\ CO_2}$	: Non CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ)
$N_{y,i,j}$	: Number of improved cookstoves of type i and batch j operating during year y.
0.95	: Discount factor to account for leakage.

Determination of fraction of woody biomass $F_{nr,y}$

For biomass, the fraction is 0.93 which is determined at the grouped level as per TOOL30 “Calculation of the fraction of non – renewable biomass”, version 04. The analysis was independently completed by C4 Ecosolutions (Pty) Ltd, below:

The fraction of harvested biomass that can be established as non-renewable biomass is the f_{nr} . This fraction ranges between 0 and 1, where 0 indicates that 100% of harvested biomass is renewable (0% is non-renewable) and 1 indicates that 100% is non-renewable. The following equation is used to calculate f_{NRB} :

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where:

f_{NRB} = Fraction of non-renewable biomass

NRB = Non-renewable biomass (t/yr)

RB = Renewable biomass (t/yr)

The quantity of non-renewable biomass (NRB) is the difference between the annual biomass consumption (H) and the quantity of renewable biomass (RB), calculated by the following equation:

$$NRB = H - RB$$

Where:

H = Total woody biomass consumption in the absence of project activity (t/yr)

- ***Estimating Lao PDR's woody biomass consumption***

The estimation of woody biomass consumption was derived from the FAO consumption data and World Bank population statistics, in combination with per capita wood consumption rates reported within government reports. Charcoal consumption converted to the equivalent wood biomass by the IPCC default factor and the wood volume converted to metric tonnes using the FAO default conversion factor/biomass conversion and expansion factor provided for the region— if given as an inventoried volume, such as industrial round wood.

The resulting non-renewable biomass (NRB) value was compared with the top-down product of average above-ground biomass (121.5 t/ha) and the average annual deforestation rate (34,500 ha/year). NRB, as calculated in this report according to the latest CDM Tool 30, is 43,174,852 t/yr, which is greater than the cross-check results based on deforestation (4,191,750 t/yr). It is expected that the cross-check estimate of biomass loss from deforestation would grossly under-estimate the actual extent of non-renewable biomass consumption as it only considers the average biomass stocks from areas that are completely deforested. Unsustainable wood harvesting resulting in forest degradation and significant reductions in biomass stocks, but not complete deforestation, are not considered in the crosscheck. The estimate of NRB is, therefore, considered applicable for this study given that the latest guidance provided by latest CDM Tool 30 was followed, domestic consumption statistics and data were sourced from official government reports, and that non-domestic consumption, sourced from the FAO Forestry Production and Trade database, was conservatively included.

Table 1. Population, woody biomass consumption (t/yr) and per capita consumption estimates for Lao PDR.

Variable	Value	Source
Population		
Urban	2,640,299	The World Bank
Rural	4,635,257	The World Bank
Domestic fuelwood consumption based on population and per capita consumption		
Urban (t/yr)	3,378,309	Calculated
Rural (t/yr)	6,144,602	Calculated
Non-domestic wood consumption (FAO)		
Fuelwood consumption (m ³ /yr)	9,645,749	FAO
Fuelwood Consumption (t/yr)	11,278,775	FAO
Charcoal consumption (t/yr)	975,216	FAO
Per capita wood biomass consumption		
Urban charcoal (t/capita/yr)	0.2	Survey
Urban wood (t/capita/yr)	0.41	Survey
Rural charcoal (t/capita/yr)	0.17	Survey
Rural wood (t/capita/yr)	0.75	Survey
Share of households consuming woodfuel		
Urban charcoal (%)	89	MEM, Lao PDR
Urban wood (%)	53	MEM, Lao PDR
Rural charcoal (%)	71	MEM, Lao PDR
Rural wood (%)	78	MEM, Lao PDR
Conversion factors		
Wood density (t/m ³)	0.725	FAO
Biomass conversion and expansion factor (t/m ³)	2.050	FAO
Charcoal to wood biomass factor	6	IPCC
Total woody biomass consumption (t/yr)	46,426,767	Calculated

Renewable biomass is the product of the mean annual increment of woody biomass growth and the total extent of the forest and other wooded areas of the country where wood extraction is not prohibited or geographically remote, calculated by the following equation:

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i}))$$

$$+ \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

Where:

MAI = Mean annual increment of woody biomass (t/ha/yr)

F_{forest} = Extent of forest in the relevant period (ha)

F_{other} = Extent of other land in the relevant period (ha)

P_{forest} = Extent of non-accessible area within forest areas (ha)

P_{other} = Extent of non-accessible area within other land areas (ha) i = Sub-category i of forest areas and other land areas.

Table 4.2. Total 2018 population, woody biomass consumption and per capita consumption estimates for the Lao PDR

Ecological Zone	Total forest cover (ha)	Protected cover (ha)	Remote cover (ha)	MAI (t/ha/yr)	Renewable biomass (t/yr)
Tropical dry forest	1,569,162	178,098	843,461	1.72	941,570
Tropical moist forest	3,977,593	814,101	2,316,261	0.75	634,755
Tropical mountain system	3,047,209	715,011	1,453,412	0.33	286,891
Tropical rainforest	4,484,130	1,408,731	2,146,070	1.49	1,388,699
Total	13,078,094	3,115,941	6,759,204	-	3,251,916

The difference between woody biomass consumption and renewable biomass is considered to be non-renewable Non-renewable biomass (NRB) utilisation in Lao PDR is, therefore, estimated to be 43,174,852 t/yr (Table 4.3).

The fraction of non-renewable biomass is the quotient of the non-renewable and the total biomass). The fraction of non-renewable biomass (fNRB) for Lao PDR is, therefore, estimated to be 0.93 (Table 4.3).

Table 4.3. Summary of the fraction of non-renewable biomass (fNRB) calculation for the Lao PDR.

Variable	Value	Source
Total woody biomass consumption (t/yr)	46,426,767	Table 1
Renewable biomass (t/yr)	3,251,916	Table 2
Non-renewable biomass (t/yr)	43,174,852	Calculated

Fraction of non-renewable biomass	0.93	Calculated
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The resulting fNRB value is higher than the expected fNRB values determined using the WISDOM method, as reported by Bailis et al., 2015¹⁰ for the same region. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate, given the different approaches of the methodologies, the following factors can contribute to variations in the fNRB estimates between the two methods:

- More recent population statistics (2020) for Laos were used in the present study. The more recent data represents an increase in population numbers since the 2009 data utilised by Bailis et al., 2015.
- Updated 2019 FAO forest products statistics were used in the present study, whereas the study conducted by Bailis et al., 2015 used 2013 FAO forest products statistics.
- Different methods are used to determine the accessibility which can result in significant differences in the estimated renewable biomass and consequently the fNRB value.
- Mean annual increments (MAI) were calculated using a weighted average based on the forest area of three categories (i.e. primary forests, above 20-year secondary forest, below 20-year secondary forest). Data from the 2019 Refinement of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for 'Above-ground biomass growth rates for different ecological zones' was utilised. As per the Bailis et al., 2015 study, MAI values were derived from a combination of field observations and IPCC values, followed by an estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs compared to using only IPCC values and may lead to lower estimations of non-renewable biomass (NRB).

The value is calculated as 0.93, however the uncertainty deduction for fnrb of 0.26 is applied as per the latest version of methodology, So, the updated fnrb = $0.93 \times (1 - 0.26) = 0.6882$ is applied for ER calculations as a conservative approach.

Determination of Quantity of woody biomass $B_{y,savings,i,j}$

$B_{y,savings,i,j}$ is calculated using the following formula in accordance with equation 4 of the approved methodology VMR0006 Version 1.1:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times (\eta_{new,i,j} / \eta_{old} - 1)$$

Where:

¹⁰) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272

$B_{y=1,new,i,j,survey}$: Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

η_{old} : Efficiency of baseline cookstove

$\eta_{new,i,j}$: Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y .

Alternatively, efficiency may be determined using equation 5 of methodology.

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where:

η_p : Efficiency of project stove(fraction) at start of project activity

$(DF_n)^{y-1}$: Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 : Adjustment factor to account for uncertainty related to project cookstove efficiency test.

$$\eta_p = 0.3844$$

$$DF_n = 0.99$$

Example of calculation

If $y=2$

$$\eta_{new,i,j} = 0.3844 \times 0.99^{2-1} \times 0.94 = 0.3577$$

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,i,j}$
1	0.3613
2	0.3577
3	0.3541
4	0.3506

5	0.3471
6	0.3436
7	0.3402
8	0.3368
9	0.3334
10	0.3301

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y.

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times (\eta_{new,i,j} / \eta_{old} - 1).$$

$B_{y=1,new,i,survey}$ is assumed as 5 kg/device/day, equivalent 1.825 tonnes/device/year

$$\eta_{old} = 10\%$$

Example calculation:

If $y = 2$

$$B_{y,savings,i,j} = 1.825 \times [(0.3577/0.1)-1] = 4.7034 \text{ tonnes}$$

Year	$\eta_{new,i,j}$	η_{old}	$B_{y=1,new,i,survey}$	$B_{y,savings,i,j}$
1	0.3613	10%	1.825	4.7694
2	0.3577	10%	1.825	4.7034
3	0.3541	10%	1.825	4.6382
4	0.3506	10%	1.825	4.5735
5	0.3471	10%	1.825	4.5095
6	0.3436	10%	1.825	4.4462
7	0.3402	10%	1.825	4.3835
8	0.3368	10%	1.825	4.3214
9	0.3334	10%	1.825	4.2599
10	0.3301	10%	1.825	4.1991

Determination of emission reduction by ICS of year 1 during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95$$

where

$$N_{y,i,j} = 600,000$$

$$NCV_{\text{Wood fuel}} = 0.0156 \text{ TJ/tonne}$$

$$F_{\text{NRB},y} = 0.6882$$

$$EF_{\text{wf},\text{CO}_2} + EF_{\text{wf},\text{non CO}_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation

If $y=2$

$$ER_{y,i,j} = 4.7034 \times 0.0156 \times 0.6882 \times 138.23 \times 600,000 \times 0.95 = 3,978,615 \text{ tonne per annum}$$

Year	$B_{y,\text{savings},i,j}$	$NCV_{\text{wood fuel}}$	$f_{\text{nr},y}$	$EF_{\text{wf},\text{CO}_2} + EF_{\text{wf},\text{non CO}_2}$	$N_{y,i,j}$	L_y	$ER_{y,i,j}$
1	4.7694	0.0156	0.6882	138.23	300,000	0.95	2,017,198
2	4.7034	0.0156	0.6882	138.23	600,000	0.95	3,978,615
3	4.6382	0.0156	0.6882	138.23	600,000	0.95	3,923,391
4	4.5735	0.0156	0.6882	138.23	600,000	0.95	3,868,719
5	4.5095	0.0156	0.6882	138.23	600,000	0.95	3,814,595
6	4.4462	0.0156	0.6882	138.23	600,000	0.95	3,761,011
7	4.3835	0.0156	0.6882	138.23	600,000	0.95	3,707,963
8	4.3214	0.0156	0.6882	138.23	600,000	0.95	3,655,446
9	4.2599	0.0156	0.6882	138.23	600,000	0.95	3,603,454
10	4.1991	0.0156	0.6882	138.23	600,000	0.95	3,551,982
Total						35,882,374	
Number of crediting period						10	
Average annual ERs						3,588,237	

Annual emission reduction is estimated to be (emission reductions) 3,588,237 tCO₂e/annum.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 1(from 10/03/2023 to 09/03/2024)	2,017,198	0	0	2,017,198

Year 2 (from 10/03/2024 to 09/03/2025)	3,978,615	0	0	3,978,615
Year 3 (from 10/03/2025 to 09/03/2026)	3,923,391	0	0	3,923,391
Year 4 (from 10/03/2026 to 09/03/2027)	3,868,719	0	0	3,868,719
Year 5 (from 10/03/2027 to 09/03/2028)	3,814,595	0	0	3,814,595
Year 6 (from 10/03/2028 to 09/03/2029)	3,761,011	0	0	3,761,011
Year 7 (from 10/03/2029 to 09/03/2030)	3,707,963	0	0	3,707,963
Year 8 (from 10/03/2030 to 09/03/2031)	3,655,446	0	0	3,655,446
Year 9 (from 10/03/2031 to 09/03/2032)	3,603,454	0	0	3,603,454
Year 10 (from 10/03/2032 to 09/03/2033)	3,551,982	0	0	3,551,982
Total	35,882,374	0	0	35,882,374

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	η_{old}
Data unit	Percentage
Description	Efficiency of pre project device
Source of data	Default value as per applied methodology
Value applied	10
Justification of choice of data or description of measurement methods and procedures applied	The default value of 0.10 is used as the replaced system is a three stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, i.e. without a grate or a chimney.

Purpose of Data	Calculation of emission reduction
Comments	The value is fixed ex - ante

Data / Parameter	$f_{NRB,y}$
Data unit	%
Description	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass
Source of data	Independent report by third party
Value applied	68.82
Justification of choice of data or description of measurement methods and procedures applied	Tool30 “Calculation of fraction of non- renewable biomass” version 04.0 was used to calculate f_{NRB} values. Project proponents considered the area where biomass is sourced and justify the selection of the area in project design documents. For this project, secondary data is used to determine the parameter as per the calculation of the fraction of non-renewable biomass, Version 04. The analysis was independently completed by third party. With regards to the calculations, the third party analyze the f_{NRB} based on satellite data availability, Geographic Information System (GIS) and data modelling which provided the most reliable measuring method. The value is calculated as 0.93, however the uncertainty deduction for f_{nrB} of 0.26 is applied as per the latest version of methodology, So, the value of 0.6882 is applied for ER calculations as a conservative approach
Purpose of Data	Calculation of baseline emission
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/Tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC default for wood fuel
Value applied	0.0156

Justification of choice of data or description of measurement methods and procedures applied	IPCC Default value
Purpose of Data	Calculation of baseline emission
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario.
Source of data	2006 IPCC
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of baseline emission
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	$EF_{wf,nonCO_2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario.
Source of data	2006 IPCC
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of baseline emission
Comments	This parameter shall remain fixed for the monitoring periods

Data / Parameter	η_p
Data unit	fraction
Description	Efficiency of the project device at the start of project activity
Source of data	Manufacturer's specification
Value applied	0.3844
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex ante
Purpose of Data	Calculation of $n_{new,i,j}$
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y.
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for project activities and programme of activities.
Frequency of monitoring/recording	At least once every two years
Value applied	600,000
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	-.
Purpose of data	Calculation of baseline emissions
Calculation method	-

Comments	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value																						
Data / Parameter	$\eta_{\text{new},y,i,j}$																						
Data unit	Fraction																						
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity																						
Source of data	Calculation																						
Description of measurement methods and procedures to be applied	To adopt Option 5 given in the methodology: "Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"																						
Frequency of monitoring/recording	Annually																						
Value applied	For ex ante calculation, the value below is applied <table border="1"> <thead> <tr> <th>Year (y)</th><th>$\eta_{\text{new},i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>0.3613</td></tr> <tr><td>2</td><td>0.3577</td></tr> <tr><td>3</td><td>0.3541</td></tr> <tr><td>4</td><td>0.3506</td></tr> <tr><td>5</td><td>0.3471</td></tr> <tr><td>6</td><td>0.3436</td></tr> <tr><td>7</td><td>0.3402</td></tr> <tr><td>8</td><td>0.3368</td></tr> <tr><td>9</td><td>0.3334</td></tr> <tr><td>10</td><td>0.3301</td></tr> </tbody> </table>	Year (y)	$\eta_{\text{new},i,j}$	1	0.3613	2	0.3577	3	0.3541	4	0.3506	5	0.3471	6	0.3436	7	0.3402	8	0.3368	9	0.3334	10	0.3301
Year (y)	$\eta_{\text{new},i,j}$																						
1	0.3613																						
2	0.3577																						
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5	0.3471																						
6	0.3436																						
7	0.3402																						
8	0.3368																						
9	0.3334																						
10	0.3301																						
Monitoring equipment	-																						
QA/QC procedures to be applied	-																						
Purpose of data	Calculation of baseline emission																						
Calculation method	-																						
Comments	Calculation to be performed using equation below:																						

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Quantity of woody biomass used by project devices in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey.
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programmes of activities and guideline for sampling and surveys for CDM project activities and programmes of activities. Determined in the first year of the introduction of the devices (e.g. During the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement measurement campaigns) may only be used if the following conditions are satisfied. (i) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; (ii) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of fire wood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as 5kg/device/day or equal to 1.825 tonnes/device/year.
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	-.

Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	

Data / Parameter	Life span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacture's specification
Description of measurement methods and procedures to be applied	PP will conduct periodic audits and surveillance of the stoves that will be distributed under the project activity to ensure their proper functioning throughout the project lifetime. This along with spot audits and after installation maintenance services, will ensure that the project stoves continue to work at efficiencies as specified by the manufacturer.
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	10
Monitoring equipment	
QA/QC procedures to be applied	
Purpose of data	For calculation of baseline emission
Calculation method	-
Comments	

5.3 Monitoring Plan

(a) Sampling Methodology

Representative sampling will be undertaken that is designed in line with the requirements of the VMR0006 methodology applied and the “Standard for sampling and surveys for CDM project activities and programme of activities”, version 09.0. This sampling plan follows the recommended outline as contained in “Guidelines for sampling and surveys for CDM project activities and programme of activities”, version 04.0.

Monitoring Parameters

Parameter	Description	Parameter type /frequency	Methods to be applied
$N_{y,i,j}$	Number of commissioned project devices of type i and batch j operating during year y	Proportion/biennially	Visits to the premises, visual inspection and interview with ICS end-user
$B_{y=1,new,l,survey}$	Quantity of woody biomass used by project devices in tonnes per device of type l and batch j	Mean value/Once, at first monitoring period. This value will be fixed for entire crediting period	Visits to the premises, visual inspection, measurement and interview with ICS end user.

To monitor the required parameters data will be collected from a statistically valid sample of ICS in the database for the monitoring period, as per this sampling plan. The method of collecting data will be field surveys and the frequency of data collection is as per the frequency defined for each parameter in the above section. Data will be collected from the field surveys, entered in the database and included in the monitoring report.

As per approved methodology, the following guidance is being used for the representative sampling method:

“A statistically valid sample of the locations where the devices are deployed, with consideration, in the sampling design, of occupancy and demographic differences can be used to determine parameter values used to calculate emission reductions, as per the relevant requirements for sampling in the “Standard for sampling and surveys for CDM

project activities and programmes of activities”. When biennial inspection is chosen a 95 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampling parameter. On the other hand, when the project proponent chooses to inspect annually, a 90 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampled parameters. In cases where survey results indicate that 90/10 precision or 95/10 precision are not achieved, the lower bound of the 90 per cent or 95 per cent confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts to achieve the 90/10 or 95/10 precision.”

Sample size will be chosen for a 95/10 precision (95% confidence interval and 10% margin of error); in cases where survey results indicate that 90/10 precision is not achieved, the lower bound of a 95% confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts to achieve a 95/10 precision.

As per “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 09, the following Sampling plan has been developed:

As per para 3 of “Standard: Sampling and Surveys for CDM project activities and programme of activities”, version 09” The general requirements of sampling methods shall be applicable to PoAs with any requirement specified in the applicable methodologies having precedence”.

Additional requirement for grouped project as per sampling standard, version 09, para 21 to 23, EB 110 will be followed:

In case a single sampling plan for more than one instance is used, “parameter values shall be estimated by sampling in accordance with the requirements in the applied methodology separately and independently for each of the instance project activity included in the grouped project except when a single sampling plan covering a grouped project is undertaken applying 95/10 confidence/precision for the sample size calculation”.

(b) Sample Frame

The sampling frame refers to all the information in the database. The grouped project is open to different instance Implementers and different models of ICS. As explained below, to take the different characteristics of different project proponent and ICS models and other factors that may affect estimates into consideration, the instances shall be either sampled separately and independently, or grouped together under a single sampling plan across instances, in each case using strata that have unique combinations of these factors.

(i) Target Population

The overall target population is the ICS distributed. The ICS to be sampled will be drawn from the list of individual ICS serial ID numbers contained in the sales record database, which is maintained by the distributor

(ii) Sampling method

Sampling will be conducted using stratified random sampling techniques, and detailed calculations are provided within the monitoring plan as per CDM guidelines “Sampling and surveys for CDM project activities and programmes of activities”. Optionally, other sampling approaches may be used in accordance with “Sampling and surveys for CDM project activities and programmes of activities” and Guideline for Sampling and Surveys for CDM Project. Activities when sampling techniques or statistical analysis necessitates it.

The improved cookstoves (ICSs) were selected randomly after arranging them in chronological order of date of distribution and assigning a sampling number to each ICS in each strata (each project activity instance). To ensure a random selection of ICS, Samples will be drawn by using the random number generator. The populations of each strata will be combined together, the sample size is determined, and a single survey will be undertaken across the project activities to collect data. To ensure the survey result is representative of the entire population, the dissimilarity (such as ICS type, vintage and project region in which they are operating) among the strata’s will be taken into account in the sample size calculation. The ICS of same type, vintage and project region in which they are operating will be grouped in the same strata.

(iii) Sample size

The sample size will be calculated using the EB 86, “Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 04.0.

According to the “Standard for sampling and surveys for CDM project activities and programme of activities”, if there is more than one parameter to be estimated, then a sample size calculation should be done for each of them. Then either the largest number for the sample size is chosen as sampling effort with one common survey, or separate sampling efforts and surveys are undertaken for each parameter.

In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per Guidelines for Sampling and surveys for CDM project activities and programmes of activities (version 04.0) Appendix 1 paragraph 5, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results;
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher’s own experiences.

For the registration/inclusion purpose of instance, option C shall be applied. Thus, there are different equations to calculate a required sample size for different situations. Different equation that will be used, depends on the type of parameter of interest, that is either:

- (a) A percentage –retention rate of improved cookstoves and continued-use rate of displaced traditional cookstoves; or
- (b) A numeric value –the mean value.

For all of the parameters 95% confidence is required that the margin of error in the estimate is not more than $\pm 10\%$ in relative terms.

For a percentage – sampling size for baseline survey

The calculation of the required sample size for parameter $N_{y,i,j}$ in the baseline survey is illustrated below for a 95/10 level of confidence and precision at grouped project level. As per Guideline “Sampling and surveys for CDM project activities and programmes of activities”, for the use of a single sampling plan covering a group of PAs, provided the homogeneity of population can be demonstrated, or differences are taken into account in the sample size calculation, a 95/10 confidence/precision is applied for annual sampling. To estimate the sample size for parameters $N_{y,i,j}$ the following equation is used.

$$n \geq \frac{1.96^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.96^2 p(1-p)}$$

Where,

- n : Sample size
- N : Population size (Total number of households/ICS)
- p : Expected proportion
- 1.96 : Represents the 95% confidence required
- 0.1 : Represents the 10% relative precision

The project activity is distribution of ICSs to households in Laos, the number of stoves expected to be distributed in this grouped project is 600,000 stoves. Hence, population size, N, is taken as 600,000 households. It is expected is 0.5, hence the expected proportion p for $N_{y,i,j}$ is taken as 0.5.

The calculation of the required sample size for each parameter in the survey is illustrated below for a 95/10 level of confidence and precision. In all cases a conservative approach is taken, however if for any parameter the required 95/10 confidence/precision is not met then the CME will randomly select an additional sample and collect further data from this sample to ensure the pooled data meet or exceed the required thresholds.

Based on the above assumptions, the resulting sampling size for a 95/10 confidence/precision is calculated as:

$$n \geq \frac{1.96^2 \times 600,000 \times 0.5(1 - 0.5)}{(600,000 - 1) \times 0.1^2 \times 0.5^2 + 1.96^2 \times 0.5(1 - 0.5)}$$

Which comes out to be

$$n \geq 383.9$$

$$n = 384$$

Therefore, the required sample size is at least 384 households. We expect the response rate from the sampled households to be 90%. Thus, we would decide to sample $384/0.9 = 430$ samples

For a percentage –retention rate of improved cookstoves, the following equation will be used

For this parameter, the monitoring survey will be conducted at grouped project level, so 95/10 level of confidence and precision will be applied for monitoring period. The project activity is distribution of ICSs to households in Laos, the number of stoves expected to be distributed in this grouped project is 600,000 stoves. Hence, population size, N, is taken as 600,000 households

$$n \geq \frac{1.96^2 N \times p(1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.96^2 p(1 - p)}$$

Where:

N	: Sample size
N=N _y	: Number of ICS in the instance project activity installed in year y = total number of households.
p	: Our expected proportion ([proportion] assuming [proportion] of the installed (ICS) will be in use and operating at the specified efficiency in first year)
1.96	: Represents the 95 % confidence required
0.1	: Represents the 10 % relative precision

Substituting the values in above equation gives the value of n = [Number of users to be sampled] (rounded up).

The following assumptions are made to exemplify the sample size calculation for parameters: N_{y,i,j}

The project proponents will distribute 600,000 ICS under grouped project activity. Hence, population size, N, is taken as 600,000 households/ICS.

It is expected at least 90% of ICS will be operational, hence the expected proportion p for $N_{y,i,j}$ is taken as 0.9.

Based on the above assumption, the resulting sampling size is calculated as:

$$n \geq \frac{1.96^2 \times 600,000 \times 0.9 (1-0.9)}{(600,000 - 1) \times 0.1^2 \times 0.9^2 + 1.96^2 \times 0.9 (1 - 0.9)}$$

Which comes out to be

$$n \geq 42.68$$

$$n = 43$$

Therefore, in this case a sample size of 43 is to be sampled from each primary sampling unit.

In case the resulting sample size to achieve the desired confidence/precision levels is smaller than 30 ICS, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion. The sample size is 43.

For a numeric value –the mean value for first monitoring period.

For the purposes of determining sample size in the first monitoring period, the efficiency of stove will vary.

For this parameter, the monitoring survey will be conducted at grouped project level, so 95/10 level of confidence and precision will be applied for monitoring period. The project activity is distribution of ICSs to households in Laos, the number of stoves expected to be distributed in this grouped project is 600,000 stoves. Hence, population size, N, is taken as 600,000 households

To estimate the sample size for parameter $B_{y=1,new,i,survey}$, the following equation is used:

$$n \geq \frac{1.96^2 NV}{(N - 1) \times 0.1^2 + 1.96^2 \times V}$$

Where:

$$V = (SD/mean)^2$$

n = Sample size

N = Population size (Total number of households/ICS)

Mean= Expected mean of ICS thermal efficiency in first year.

SD = Expected standard deviation

1.96 = Represents the 95% confidence required

(In the case of 90% confidence, 1.645 shall be used)

0.1 = Represents the 10% relative precision

The expected mean of ICS thermal efficiency in first year is 0.3613 and its standard deviation is 0.03613.

Based on the above assumptions, the sample size calculation for a 95/10 confidence/precision would be

$$n \geq \frac{1.96^2 \times 600,000 \times (0.03613/0.3613)^2}{(600,000 - 1) \times 0.1^2 + 1.96^2 \times (0.03613/0.3613)}$$

Using above values, the sample size comes out to be 3.84 or 4.

As the calculated sample size (for parameter of interest is a numeric mean value) is less than 30 samples, since $B_{y=1, \text{new}, i, \text{survey}}$ is numeric mean value, the Student's t-distribution will be used to calculate the sample size (in accordance with paragraph 14 of standard: sampling and survey for CDM project activities and programmes of activities (version 09.0)).

Monitoring Parameter	$B_{y=1, \text{new}, i, \text{survey}}$				
Sampling approach	Stratified random sampling based on stove year of distribution				
Stratum Number	Year of Distribution	Population Size	Expected Mean	Expected Standard Deviation	Calculated Sample Size
Year 2023	2023	600,000	0.3613	0.03613	7
Sample size determination					
Mean					0.3613

Standard Deviation (SD)		0.03613
$V_{\text{mean}} = (\text{SD}/\text{mean})^2$		0.01000
Minimum Sample Size required		4
t-Distribution sample size adjustment	Iteration 1	11
	Iteration 2	5
	Iteration 3	8
	Iteration 4	6
	Iteration 5	7
	Iteration 6	6

The adjusted sample size obtained after applying Student's t-distribution is 7. Therefore, in this case a sample size of 7 is to be sampled from each primary sampling unit.

Since parameters $N_{y,j,j}$ and $B_{y=1,\text{new},i, \text{survey}}$ share the same sampling units, CME may choose to have one common survey for these two parameters with largest number of sample size between these two parameters being chosen. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

The parameter $B_{y=1,\text{new},i, \text{survey}}$ only monitored in first monitoring period and will be default value for later monitoring period. Sampling procedure for this parameter only applied in first monitoring period.

Oversampling is strongly encouraged, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved and additional sampling efforts would be required. The sample size shown above will be adjusted upwards to account for non-responses, project proponent shall determine the appropriate non-responses rate based on previous experience.

(iv) Survey and Data collection Method

The sample to be surveyed will be drawn randomly with a geographical spread within the project boundary of the project activity instance

A survey will be conducted through: physical on-site visit (face-to-face). For the physical on-site visit (face-to-face interview), data is collected through hard-copy questionnaires.

Training

The project proponent will conduct a training programme for project participant's representative and its nominated distributor or third party surveyor for survey framework and approach. Major focus will be on understanding collection of data as per the survey questionnaire.

Data Collection

The survey team will collect the data as per the sampling method.

Data entry and storage

Level of supervision and guidance provided to staff, for example written instructions; The standardized system used to initially enter data into a computer and produce a final dataset, documenting steps taken to minimize the introduction of errors. There will be Systems in place to ensure all collected data is processed.

The type of quality checks performed on data entered, for example range checks, inconsistency checks, checking of subsamples of data by supervisors will be done. Clear guidelines will be done on How errors are recorded, reported and the remedial action taken. Level of security and type of backup processes to guarantee data integrity, for example methods to prevent fraud and accidental deletion will be done.

Data Analysis

The data will be analysed for consolidation of findings.

The data from the survey will be analyzed.

- Determine the mean value of quantity of woody biomass used by project devices in first monitoring period.
- To determine number of project ICS that are still in operation

Report generation.

A report will be prepared. The report should contain explicit statements, explanations or content related to the following: (a) A suitable qualification of the data analyst; (b) The statistical design of the survey, justification of its choice and the sample size used; (c) The statistical methods employed to analyse the data and reasons for choosing these methods, which should cite official or industry guidelines or other references where appropriate.

Quality Assurance/Quality Control:

The overall monitoring and the implementation of the sampling plan will be co-ordinated by the project proponent. They will ensure successful monitoring of the emission reductions of the proposed project during its crediting period. Furthermore, the survey of the representative sample for the parameters will be carried out by the project proponent in conjunction with the project proponent.

During the survey, in order to anticipate any low response rate and answers bias, 10% will be applied over the determined sample size. The survey team will ensure that all questionnaires are. To remedy the incomplete questionnaires, additional households/users will be drawn randomly until the required number is met at per the sample size determined. There might be changes of getting outliers while sampling. The following approach will be used to identify and address outliers for the samples during monitoring. Outliers will be defined as those data points with values greater than three standard deviations from the mean of the sample.

Data points identified as outliers according to the above analysis will be examined further to correct for possible transcription and data entry errors, but will be omitted from the analysis if no such administrative errors exist.

Analysis

The data from the survey will be analyzed

Determine the mean value of the efficiency of the ICS that are still operating in first monitoring period.

To determine number of project ICS that are still in operation

Implementation plan

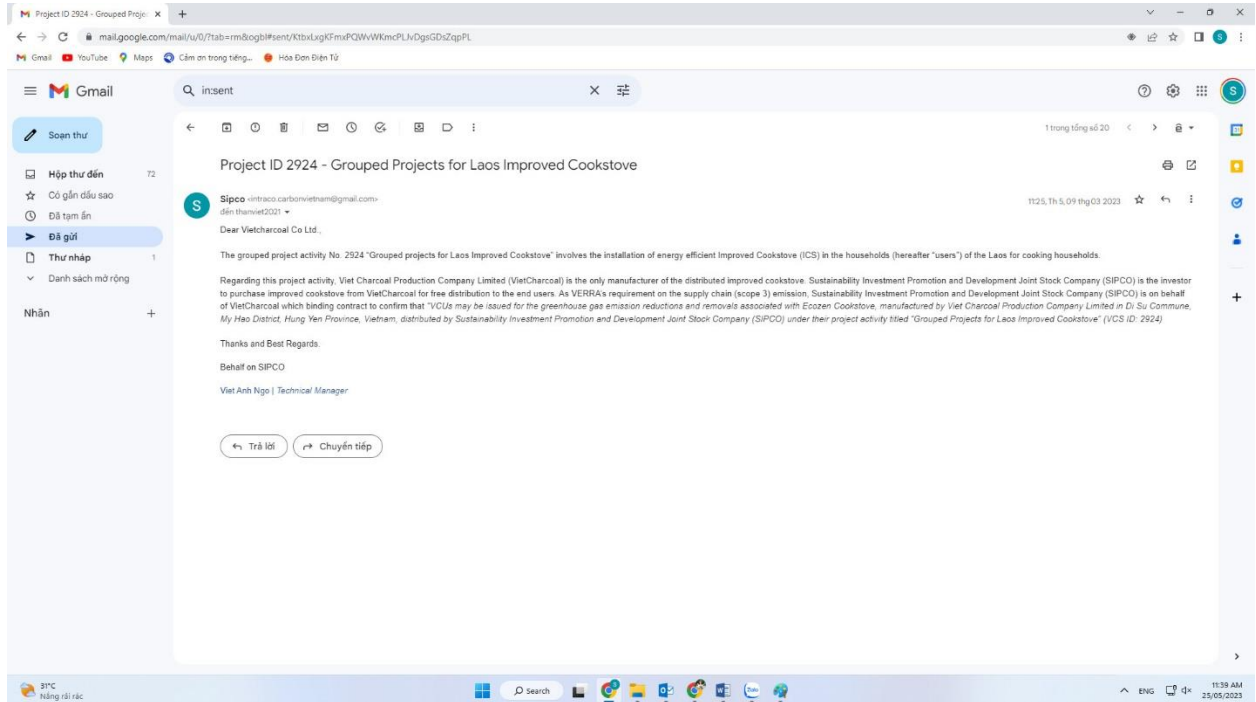
Survey Implementation Schedule:

The survey for collecting the details of the number of ICS in operation will be conducted at least every 2 years and will be scheduled during the last month of the monitoring period. If the time period between surveys should exceed 2 years, then no emission reductions shall be claimed for such period.

The overall monitoring and the implementation of the sampling will be co-ordinated by the project proponent and the management staff. They will ensure successful monitoring of the emission reductions of the proposed project during its crediting period. Furthermore, the survey of the representative sample for the parameters will be carried out by the distributor in conjecture with the project proponent.

APPENDIX 1: SUPPLY CHAIN

1. Email SIPCO sent to Viet Charcoal about supply chain Scope 3



2. Public statement has been posted on SIPCO's website:

<https://intracarbon.com/public-announcement-on-grouped-projects-for-laos-cookstove-program-project-id-2924>

